3. Stopping Primitive

Define a primitive function to **STOP** the vehicle within selected **boundaries**.

- **Final** Conditions:
  $a_f = 0 \;\; \rightarrow$ final acceleration equal to zero;
  $v_f = 0 \;\; \rightarrow$ final velocity equal to zero;
  $s_f = \bar{s}_f \;\; \rightarrow$ final position equal to the position of the traffic light;
- **Initial** Conditions:
  $v_0 = \bar{v}_0 \rightarrow$ initial velocity given by the sensor;
  $a_0 = \bar{a}_0 \rightarrow$ initial acceleration given by the sensor;

Find the optimal **time** solution:
the total cost is derived w.r.t. t_f and imposed equal to zero $\rightarrow$ new function to define the optimal **time** needed to reach a given **final position** (traffic light).

3. Stopping Primitive